

pharmaniaga®

Labetalol Hydrochloride

25mg/5mL Injection



COMPOSITION

Each 5 ml contain Labetalol hydrochloride 25mg. It also contains the following excipients:

- Dextrose Anhydrous
- Disodium Edetate
- Citric Acid Anhydrous
- Water for Injection

DESCRIPTION

Clear, colourless to pale yellow. Pharmaniaga Labetalol Hydrochloride 25mg/5ml Injection is compatible with the following infusion solutions:

- 5% Dextrose BP
- 0.18% Sodium Chloride and 4% Dextrose BP
- Compound Sodium Lactate BP

PHARMACODYNAMICS

Pharmacotherapeutic group: Alpha and beta blocking agents

ATC Code: C07AG01

Mechanism of Action

Labetalol lowers the blood pressure primarily by blocking peripheral arteriolar alpha-adrenoceptors, thus reducing peripheral resistance, and by concurrent beta-blockade, protects the heart from reflex sympathetic drive that would otherwise occur. Cardiac output is not significantly reduced at rest or after moderate exercise. Increases in systolic blood pressure during exercise are reduced but corresponding changes in diastolic pressure are essentially normal.

In patients with angina pectoris coexisting with hypertension, the reduced peripheral resistance decreases myocardial afterload and oxygen demand. All these effects would be expected to benefit hypertensive patients and those with coexisting angina.

PHARMACOKINETICS

The plasma half-life of labetalol is about four hours. About 50% of Labetalol in the blood is protein bound. Labetalol is

metabolized mainly through conjugation to inactive glucuronide metabolites. These are excreted both in urine and via the bile, into the faeces. Only negligible amounts of Labetalol cross the blood brain barrier in animal studies. Labetalol crosses the placental barrier and secreted in breast milk.

INDICATION

- Severe hypertension, including severe hypertension of pregnancy, when rapid control of blood pressure is essential
- Anaesthesia when a hypotensive technique is indicated
- Hypertensive episode following acute myocardial infarction

RECOMMENDED DOSAGE

Labetalol hydrochloride injection is intended for i.v. use in hospitalised patients. The plasma concentration achieved after intravenous dose of Labetalol in severe hypertension are substantially greater than those following oral administration of the drug and provide a greater degree of blockade of alpha-adrenoceptors necessary to control the more severe disease. Patients should always receive the drug whilst in the supine or left lateral position. Raising the patient into the upright position within 3 h of i.v. labetalol hydrochloride administration should be avoided since excessive postural hypotension may occur. It is desirable to monitor the blood pressure and heart rate injection and during infusion. In most patients, there is a small decrease in the heart rate; severe bradycardia is unusual but may be controlled by injecting atropine 1 to 2mg intravenously. Respiratory function should be observed particularly in patients with any known impairment.

Once the blood pressure has been adequately reduced by bolus injection or infusion, maintenance therapy with labetalol hydrochloride tablets should be substituted with a starting dose of 100 mcg twice daily. Labetalol hydrochloride injection has been administered to patients with uncontrolled hypertension already receiving other hypotensive agents, including beta-blocking drugs, without adverse effects.

Adults

For severe Hypertension

Bolus injection

If it is essential to reduce blood pressure quickly, as for example, in hypertensive encephalopathy, a dose of 50mg of labetalol hydrochloride should be given by intravenous injection over a period of at least 1 minute. If necessary, doses of 50mg may be repeated at 5 minutes intervals until a satisfactory response occurs. The total dose should not exceed 200mg. After bolus injection, the maximum effect usually occurs within five minutes and the effective duration of action is usually about 6 hours but may be as long as 18 hours.

Intravenous Infusion

A 1 mg/ml solution of Labetalol hydrochloride injection should be used, i.e. the content of two 20 ml vials or eight 5ml ampoules (200 mg) diluted to 200 ml with Sodium Chloride and Dextrose Injection BP or 5% Dextrose Intravenous Infusion BP.

- Severe Hypertension of Pregnancy

Infusion should be started at 20 mg/h, then doubled every 30 min until a satisfactory response is obtained or a dosage of 160 mg/h is reached. Occasionally higher doses may be necessary.

- Hypertension Due to Other Causes

Infusion should be started at a rate of 2 mg/min until a satisfactory response is obtained, then stopped. The effective dose is usually 50 to 200 mg but larger doses may

be needed, especially in patients with phaeochromocytoma. The rate of infusion may be adjusted according to the response at the discretion of the physician.

For Hypotensive Anaesthesia

Induction should be with standard agents (e.g. sodium thiopentone) and anaesthesia maintained with nitrous oxide and oxygen with or without halothane. The recommended starting dose of labetalol hydrochloride is 10 to 20 mg intravenously depending on the age and condition of the patient. Patient for whom halothane is contraindicated usually require a higher initial dose of labetalol hydrochloride (25 to 30 mg). If satisfactory hypotension is not achieved after 5 min, increment of 5 to 10 mg should be given until the desired level of blood pressure is attained.

Halothane and labetalol hydrochloride act as synergistically therefore the halothane concentration should not exceed 1 to 1.5% as profound fall in blood pressure may be precipitated. Following labetalol hydrochloride injection the blood pressure can be quickly and easily adjusted by altering the halothane concentration and/or adjusting table tilt. The mean duration of hypotension following 20 to 25 mg of i.v. labetalol hydrochloride is 50 min. Hypotension induced by labetalol hydrochloride injection is readily reversed by atropine 0.6mg and discontinuation of halothane.

Tubocurarine and pancuronium may be used when assisted or controlled ventilation is required. Intermittent positive pressure ventilation may further increase the hypotension resulting from labetalol hydrochloride injection and/or halothane.

For Hypertensive Episode Following Acute Myocardial Infarction

Infusion should be started at 15 mg/h and gradually increased to a maximum of 120 mg/h depending on the control of blood pressure.

Children

Safety and efficacy have not been established.

ROUTE OF ADMINISTRATION

Intravenous infusion.

CONTRAINDICATIONS

Labetalol Hydrochloride injection is contraindicated in second or third-degree heart block, cardiogenic shock and other conditions associated with severe and prolonged hypotension or severe bradycardia. Beta-blockers, even those with apparent cardioselectivity, should not be used in patients with asthma or a history of obstructive airway disease. Labetalol hydrochloride is contraindicated for patients known to have hypersensitivity to the drug. When peripheral vasoconstriction suggests low cardiac output, the use of labetalol hydrochloride injection to control hypertensive episodes following acute myocardial infarction is contraindicated.

WARNINGS AND PRECAUTIONS

There have been reports of skin rashes and/or dry eyes associated with the use of beta-adrenoceptor blocking drugs. The reported incidence is small and in most cases the symptoms have cleared when the treatment was withdrawn. Gradual discontinuance of the drug should be considered if any such reaction is not otherwise explicable. The occurrence of intraoperative floppy iris syndrome (IFIS, a variation of Homer's syndrome) has been observed during cataract surgeries in some patients who were being treated with tamsulosine, or have been treated with tamsulosine in the past. IFIS has also been reported when other alpha-1-blockers were being used, and the possibility of a class effect cannot be excluded. Since IFIS can lead to a

higher chance of complications during cataract surgeries, theophthalmologist needs to be informed if alpha-1-blockers are currently being used, or have been used in the past. There have been rare reports of severe hepatocellular injury with labetalol therapy. The hepatic injury is usually reversible and has occurred after both short and long term treatment. Appropriate laboratory testing should be done at the first sign or symptom of liver dysfunction. If there is laboratory evidence of liver injury or the patient is jaundiced, labetalol therapy should be stopped and not re-started.

Due to negative inotropic effects, special care should be taken with patients whose cardiac reserve is poor and heart failure should be controlled before starting labetalol therapy.

Patients, particularly those with ischemic heart disease, should not interrupt/discontinue abruptly labetalol therapy. The dosage should gradually be reduced, i.e. over 1-2 weeks, if necessary at the same time initiating replacement therapy, to prevent exacerbation of angina pectoris. In addition, hypertension and arrhythmias may develop.

It is not necessary to discontinue labetalol therapy in patients requiring anaesthesia, but the anaesthetist must be informed and the patient should be given intravenous atropine prior to induction. During anaesthesia labetalol may mask the compensatory physiological responses to sudden haemorrhage (tachycardia and vasoconstriction). Close attention must therefore be paid to blood loss and the blood volume maintained. If beta-blockade is interrupted in preparation for surgery, therapy should be discontinued for at least 24 hours. Anaesthetic agents causing myocardial depression (e.g. cyclopropane, trichloroethylene) should be avoided. Labetalol may enhance the hypotensive effects of halothane.

In patients with peripheral circulatory disorders (Raynaud's disease or syndrome, intermittent claudication), beta-blockers should be used with great caution as aggravation of these disorders may occur.

Beta-blockers may induce bradycardia. If the pulse rate decreases to less than 50-55 beats per minute at rest and the patient experiences symptoms related to the bradycardia, the dosage should be reduced.

Beta-blockers, even those with apparent cardio selectivity, should not be used in patients with asthma or a history of obstructive airways disease unless no alternative treatment is available. In such cases the risk of inducing bronchospasm should be appreciated and appropriate precautions taken. If bronchospasm should occur after the use of labetalol it can be treated with a beta2-agonist by inhalation, e.g. salbutamol (the dose of which may need to be greater than the usual in asthma) and, if necessary, intravenous atropine 1mg.

Due to a negative effect on conduction time, beta-blockers should only be given with caution to patients with first degree heart block. Patients with liver or kidney insufficiency may need a lower dosage, depending on the pharmacokinetic profile of the compound. The elderly should be treated with caution, starting with a lower dosage but tolerance is usually good in the elderly.

Patients with a history of psoriasis should take beta-blockers only after careful consideration.

Risk of anaphylactic reaction: While taking beta-blockers, patients with a history of severe anaphylactic reaction to a variety of allergens may be more reactive to repeated

challenge, either accidental, diagnostic or therapeutic. Such patients may be unresponsive to the usual doses of epinephrine used to treat allergic reaction.

INTERACTIONS WITH OTHER MEDICAMENTS

Concomitant use not recommended:

- Calcium antagonists such as verapamil and to a lesser extent diltiazem have a negative influence on contractility and atria-ventricular conduction.

- Digitalis glycosides used in association with beta-blockers may increase atria-ventricular conduction time.

- Clonidine: Beta-blockers increase the risk of rebound hypertension. When clonidine is used in conjunction with non-selective beta-blockers, such as propranolol, treatment with clonidine should be continued for some time after treatment with the beta-blocker has been discontinued.

- Monoamineoxidase inhibitors (except MOA-B inhibitors).

Use with caution:

- Class I antiarrhythmic agents (e.g. disopyramide, quinidine) and amiodarone may have potentiating effects on atrial conduction time and induce negative inotropic effect.

- Insulin and oral antidiabetic drugs may intensify the blood sugar lowering effect, especially of non-selective beta-blockers. Beta-blockade may prevent the appearance of signs of hypoglycaemia (tachycardia).

- Anaesthetic drugs may cause attenuation of reflex tachycardia and increase the risk of hypotension. Continuation of beta-blockade reduces the risk of arrhythmia during induction and intubation. The anaesthesiologist should be informed when the patient is receiving a beta-blocking agent. Anaesthetic agents causing myocardial depression, such as cyclopropane and trichlorethylene, are best avoided.

- Cimetidine, hydralazine and alcohol may increase the bioavailability of labetalol.

- Several different drugs or drug classes may enhance the hypotensive effects of labetalol: ACE inhibitors; angiotensin-11 antagonists; aldesleukin, alprostadil; anxiolytics; hypnotics; moxislyte; diuretics; alpha-blockers.

- Several different drugs or drug classes may antagonize the hypotensive effects of labetalol: NSAIDs, corticosteroids; oestrogens; progesterones.

Take into account:

- Calcium antagonists: dihydropyridine derivatives such as nifedipine. The risk of hypotension may be increased. In patients with latent cardiac insufficiency, treatment with beta-blockers may lead to cardiac failure.

- Prostaglandin synthetase inhibiting drugs may decrease the hypotensive effect of beta-blockers.

- Sympathicomimetic agents may counteract the effect of beta-adrenergic blocking agents.

- Concomitant use of tricyclic antidepressants, barbiturates, phenothiazines or other antihypertensive agents may increase the blood pressure lowering effect of labetalol. Concomitant use of tricyclic antidepressants may increase the incidence of tremor.

- Labetalol has been shown to reduce the uptake of radioisotopes of metaiodobenzylguanidine (MIBG), and may increase the likelihood of a false negative study. Care should therefore be taken in interpreting results from MIBG scintigraphy. Consideration should be given to withdrawing labetalol for several days at least before MIBG scintigraphy, and substituting other beta or alpha-blocking drugs.

- Antimalarials such as mefloquine or quinine may increase the risk of bradycardia.

- Ergot derivatives may increase the risk of peripheral vasoconstriction.

- Tropisetron may increase the risk of ventricular arrhythmia.

- Labetalol interferes with laboratory tests for catecholamines

STATEMENT ON USAGE DURING PREGNANCY AND LACTATION

Although no teratogenic effects have been demonstrated in animals, labetalol hydrochloride should only be used during the first trimester of pregnancy if the potential benefit outweighs the potential risk. In humans, labetalol crosses the placental barrier and the possibility of the consequences of alpha- and beta-adrenoceptor blockade in the fetus and neonate should be borne in mind. Perinatal and neonatal distress (bradycardia, hypotension, respiratory depression, hypoglycaemia, hypothermia) has been rarely reported. Sometimes these symptoms developed a day or two after birth. Response to supportive measure (e.g.i.v. fluids and glucose) is usually prompt but with severe preeclampsia, particularly after prolonged i.v. labetalol hydrochloride, recovery may be slower. This may be related to diminished liver metabolism in premature babies. Beta-blockers reduce placental perfusion, which may result in intrauterine foetal death, immature and premature deliveries. There is an increased risk of cardiac and pulmonary complications in the neonate in the post-natal period. Intra-uterine and neonatal deaths have been reported but other drugs (e.g. vasodilators, respiratory depressants) and the effects of pre-eclampsia, intra-uterine growth retardation and prematurity were implicated. Such clinical experience warns against unduly prolonging high dose labetalol hydrochloride and delaying delivery and against co-administration of hydralazine.

Labetalol is excreted in breast milk. Breast-feeding is therefore not recommended.

Nipple pain and Raynaud's phenomenon of the nipple have been reported.

SIDE EFFECTS/ADVERSE EFFECT

Labetalol Injection is usually well tolerated. Excessive postural hypotension may occur if patients are allowed to assume an upright position within three hours of receiving Labetalol Injection. Most side-effects are transient and occur during the first few weeks of treatment with labetalol.

They include:

Blood and the lymphatic system disorders

Rare reports of positive antinuclear antibodies unassociated with disease, hyperkalaemia, particularly in patients who may have impaired renal excretion of potassium, thrombocytopenia.

Psychiatric disorders

Depressed mood and lethargy, hallucinations, psychoses, confusion, sleep disturbances, nightmares.

Nervous system disorders

Headache, tiredness, dizziness, tremor has been reported in the treatment of hypertension of pregnancy.

Eye disorders

Impaired vision, dry eyes.

Cardiac disorders

Bradycardia, heart block, heart failure, hypotension.

Vascular disorders

Ankle oedema, increase of an existing intermittent claudication, postural hypotension, cold or cyanotic

extremities, Raynaud's phenomenon, paraesthesia of the extremities.

Respiratory, thoracic and mediastinal disorders

Bronchospasm (in patients with asthma or a history of asthma), nasal congestion, interstitial lung disease.

Gastrointestinal disorders

Epigastric pain, nausea, vomiting, diarrhoea.

Hepato-biliary disorders

Raised liver function tests, jaundice (both hepatocellular and cholestatic), hepatitis and hepatic necrosis.

Skin and subcutaneous tissue disorders

Sweating, tingling sensation in the scalp, usually transient, may occur in a few patients early in treatment, reversible lichenoid rash, systemic lupus erythematosus, exacerbation of psoriasis.

Musculoskeletal, connective tissue and bone disorders:

Cramps, toxic myopathy.

Renal and urinary disorders

Acute retention of urine, difficulty in micturition.

Reproductive system and breast disorders

Ejaculatory failure.

Frequency 'not known': Nipple pain, Raynaud's phenomenon of the nipple.

General disorders and administration site conditions

Hypersensitivity (rash, pruritus, angioedema and dyspnoea), drug fever, masking of the symptoms of thyrotoxicosis or hypoglycaemia, reversible alopecia

OVERDOSAGE

Symptoms of overdosage are bradycardia, hypotension, bronchospasm and acute cardiac insufficiency.

After an overdose or in case of hypersensitivity, the patient should be kept under close supervision and be treated in an intensive-care ward. Artificial respiration may be required. Bradycardia or extensive vagal reactions should be treated by administering atropine or methylatropine. Hypotension and shock should be treated with plasma/plasma substitutes and, if necessary, catecholamines. The betablocking effect can be counteracted by slow intravenous administration of isoprenaline hydrochloride, starting with a dose of approximately 5mcg/min, or dobutamine, starting with a dose of approximately 2.5mcg/min, until the required effect has been obtained. If this does not produce the desired effect, intravenous administration of 8-10 mg glucagon may be considered. If required the injection should be repeated within one hour, to be followed, if necessary, by an iv infusion of glucagon at 1-3 mg/hour. Administration of calcium ions, or the use of a cardiac pacemaker, may also be considered.

Oliguric renal failure has been reported after massive overdosage of labetalol orally. In one case, the use of dopamine to increase the blood pressure may have aggravated the renal failure.

Labetalol does have membrane stabilising activity which may have clinical significance in overdosage.

Haemodialysis removes less than 1% labetalol hydrochloride from the circulation.

Effect on ability to drive and use machine

There are no studies on the effect of this medicine on the ability to drive. When driving vehicles or operating machines it should be taken into account that occasionally dizziness or fatigue may occur.

INSTRUCTION FOR USE

- May be given as bolus injection by slow IV injection
- Intravenous Infusion

STORAGE CONDITIONS

Store below 30°C.

Do not freeze. Protect from light.

Retain in carton till time of use.

The solution will remain stable for 24 hours at 30°C after mixing with the compatible solutions. Unused admixtures should be discarded 24 hours after preparation.

Compatible Solutions:

Pharmaniaga Labetalol Hydrochloride 25mg/ml injection is compatible with the following i.v. infusion fluids:

- 5% Dextrose BP
- 0.18% Sodium Chloride and 4% Dextrose BP
- Compound Sodium Lactate BP

The solution should be clear, colorless when diluted. Do not use if the solution is discoloured or if there is particulate matter in the solution.

Incompatibilities:

Pharmaniaga Labetalol Hydrochloride 25mg/5ml Injection is incompatible with sodium bicarbonate injection BP 4.2% W/V.

SHELF LIFE

The injection can be used within 24 months from the date of manufacture if kept as recommended.

After diluted with compatible solution, product may be stored up to 24 hours at temperature below 30°C after mixing with the compatible solutions.

Product should not be used beyond the expiry date imprinted on the product packaging.

DOSAGE FORMS AND PACKAGING AVAILABLE

10 x 5mL ampoule (clear) of Pharmaniaga Labetalol Hydrochloride 25mg/5mL Injection.

10 ampoules per unit PVC tray were packed in paper carton.

PRODUCT REGISTRATION HOLDER/MANUFACTURER PHARMANIAGA LIFESCIENCE SDN BHD (198201002939)

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